

Opik: open-source platform for evaluating, testing and monitoring LLM applications



Opik, an open-source platform built by Comet, designed to streamline the development and deployment of LLM applications. Whether you're refining prompts, assessing model accuracy, or tracking production performance, Opik provides an end-to-end solution for maintaining the quality of your AI-driven applications.

Why Opik?

Opik is built to address key challenges in LLM application development. It offers:

1. Development Phase Support

- **Tracing:** Track all LLM calls and traces during both development and production.
- **Annotations:** Log feedback scores using Opik's Python SDK or UI.
- **Prompt Playground:** Experiment with different prompts and models effortlessly.
- **Automated Evaluation:** Store test cases, run experiments, and evaluate model performance.
- **LLM-as-a-Judge Metrics:** Leverage Opik's built-in metrics for tasks like hallucination detection, moderation, and relevance assessment.
- **CI/CD Integration:** Run evaluations within your CI/CD pipeline using Opik's PyTest integration.

2. Production Monitoring

- **Scalability:** Opik supports high-volume logging, making it suitable for applications handling millions of traces daily.
- **Monitoring Dashboards:** Visualize key metrics such as feedback scores, token usage, and trace count over time.
- **Online Evaluation Metrics:** Identify issues in production using LLM-as-a-Judge metrics.

Installing Opik

Opik is available both as a **self-hosted** open-source platform and as a **hosted solution** on Comet.com.

Getting Started with Opik

1. Install the Opik Python SDK:

```
pip install opik
```

2. Configure Opik:

```
opik configure
```

- For local deployment, use `opik.configure(use_local=True)` in your Python script.

3. Add API key

Start logging traces with Opik:

```
import opik

opik.configure(use_local=True)

@opik.track
def my_llm_function(user_question: str) -> str:
    return llm.invoke(user_question)
```

Check out different framework (langchain, llamindex) integrations [here](#).

Self-Hosting with Docker Compose

1. Clone the Opik repository:

```
git clone https://github.com/comet-ml/opik.git
```

2. Navigate to the deployment directory:
`cd opik/deployment/docker-compose`
3. Start the Opik platform:
`docker compose up --detach`
4. Access Opik UI at: **`http://localhost:5173`**

For other installation methods, check out the [deployment guides](#).

LLM-as-a-Judge Metrics

Opik offers pre-built evaluation metrics to assess LLM outputs effectively. Some key metrics include:

- **Hallucination Detection**
- **Moderation (Harmful Content Detection)**
- **Answer Relevance & Usefulness**
- **Context Recall & Precision**
- **Heuristic-Based Metrics (Regex Matching, JSON Validation, etc.)**

Example:

```
from opik.evaluation.metrics import Hallucination

metric = Hallucination()
score = metric.score(
    input="What is the capital of France?",
    output="Paris",
    context=["France is a country in Europe."]
)
print(score)
```

Check out all the metrics [here](#).

Evaluating Your LLM Application

With Opik, the evaluation doesn't stop at development:

- Store test cases and run experiments using **Datasets and Experiments**.
- Integrate evaluations into your CI/CD pipeline with **PyTest integration**.

More resources:

- [Guide](#) on How to evaluate your LLM Application with Opik.
- [Guide](#) on Datasets.
- [Guide](#) on PyTest Integration.

Conclusion

Opik simplifies the process of evaluating, testing, and monitoring LLM applications, making it an essential tool for AI developers. Whether you're in the early stages of development or managing production-scale deployments, Opik ensures reliability, transparency, and continuous improvement in your LLM workflows.